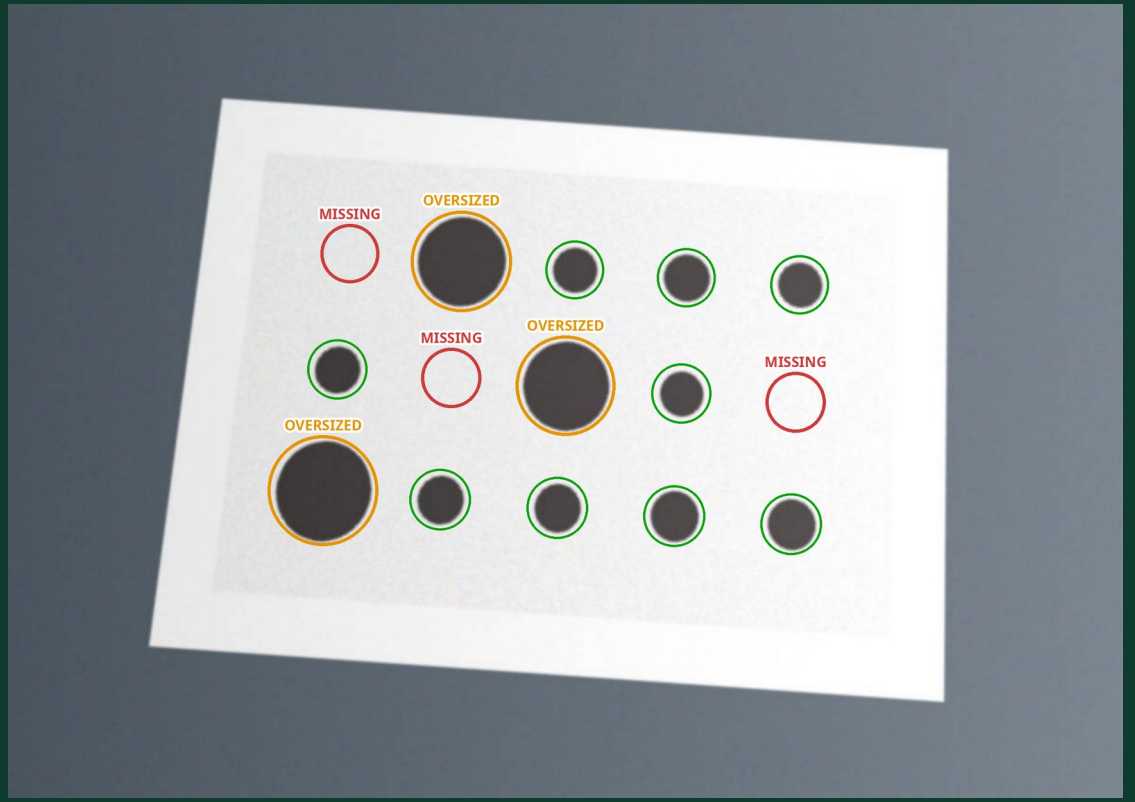
AI HORIZON 2026 · CHALLENGE 2 · NSW AUTOMATION

AI Dispensing Defect Detective

A phone photo and five questions → the likely cause, the reasons behind it, and what to check first.

And a case history that only learns from fixes that were proved.

Calvin Khoo



A phone photo of a printed sample card. Each missing deposit is followed by an oversized one: the system reads carryover, p = 0.010.

THE PROBLEM

Dispensing defects are diagnosed by experience — and experience doesn't scale

New technicians guess

Inconsistent dots have at least five plausible causes. Without the experience to tell them apart, the first check is a hunch — and a wrong strip-down costs a shift.

Three changes at once

Re-prime, raise the pressure, swap the needle: the defect goes away, and nobody can say which change fixed it. The next case starts from zero.

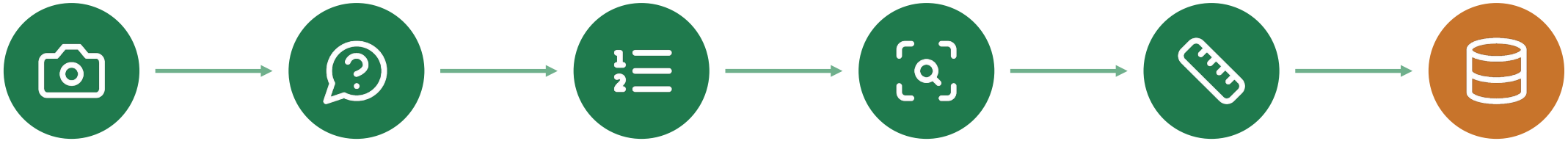
Knowledge walks out the door

What the senior engineer knows isn't recorded anywhere the next case can use it.

NSW Automation's brief: help engineers and technicians perform faster preliminary troubleshooting — not replace them.

HOW IT WORKS

One guided session, from a phone photo to a proved fix



1 Photograph

Measures size spread, shape and missing dots — however the photo is framed

2 Interview

Five questions, then follow-ups picked because their answer changes the ranking

3 Rank

Seven causes from the manual — every % built from cited reasons

4 Check

Look first, then change one thing; the ranking updates live

5 Verify

Re-photograph: in spec on every check, held 20+ shots

6 Learn

Only proved fixes become precedent for the next case

Runs offline on a laptop or line PC

No training data — classical, auditable vision

LLM optional: it can disagree, it never sets a %

THE AI

It asks what would change its mind — and shows why

A follow-up, chosen live

Watch the pressure gauge while it dispenses — does it move?

Asked because the answer can change the lead: “It wanders or dips during a run” would put pressure-time dispensing variability at 84% (now 59%).

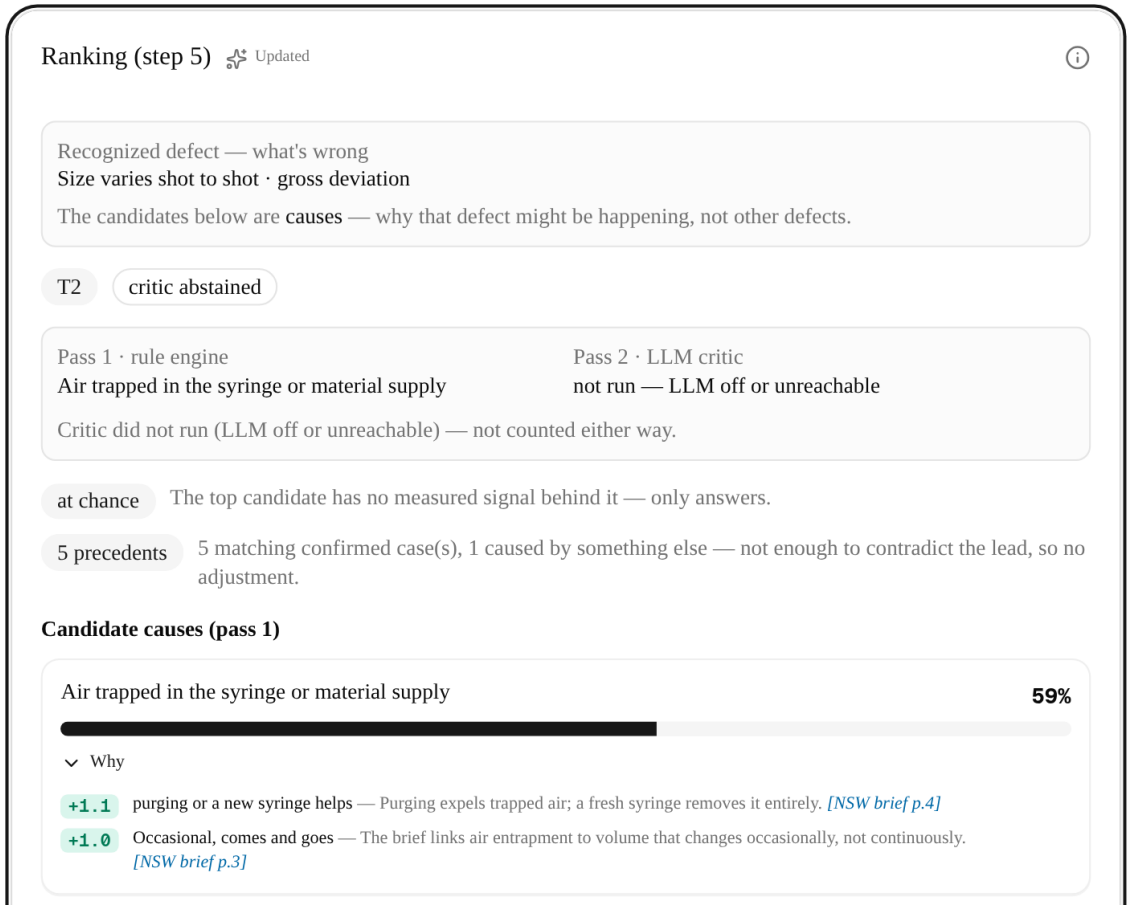
- It wanders or dips during a run — not just at start-up
- Steady at the set value — the needle doesn't move
- Can't see a gauge — adds no evidence

Every candidate question is run through the rule engine; the one asked is the one whose answer would move the ranking most.

The lead, explained in the brief's own terms

Air trapped in the syringe or material supply leads because: purging or a new syringe helps (NSW brief p.4); occasional, comes and goes (NSW brief p.3).

Tier T2 — a likely lead: Air trapped in the syringe or material supply at 59%, but it isn't backed by a strong measurement or precedent yet — see the panel on the right for why.



Each reason shows its weight and the page it came from: the NPL Best Practice Guide or the NSW brief.

THE EVIDENCE

When it says it's confident, it's right

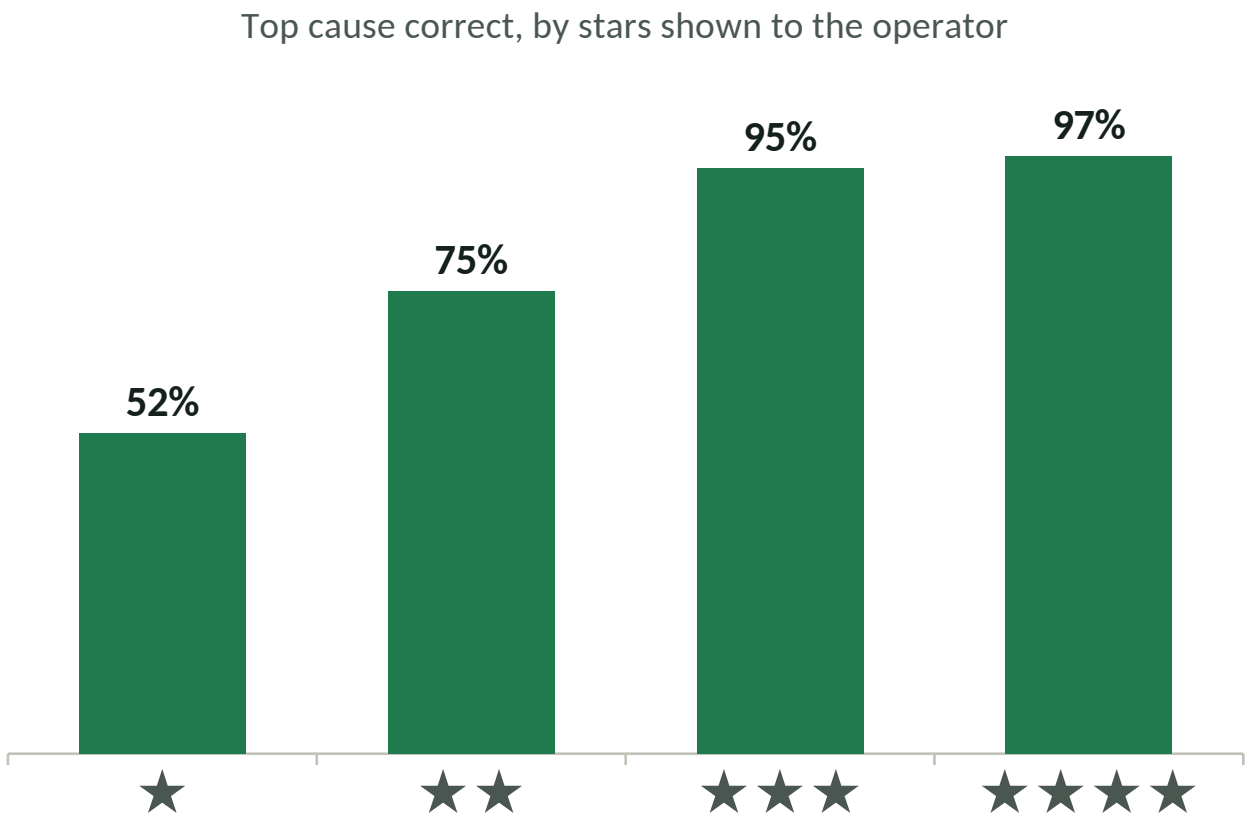


Photo + answers · 420 simulated cases at three answer-reliability levels · chance = 14%

66%

top cause right from one photo and five answers, with an operator whose answers are only 75% reliable — chance is 14%

6,600

synthetic photos and 420 simulated cases through the real code, in every evaluation run (about a minute)

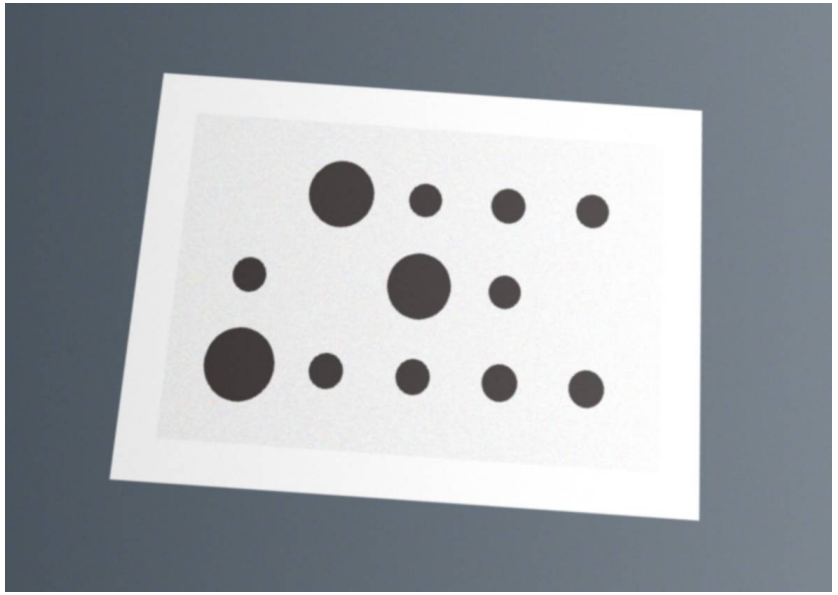
79

automated checks — 52 backend tests, 25 measurement oracles, 2 golden cases — plus a browser run of the whole wizard

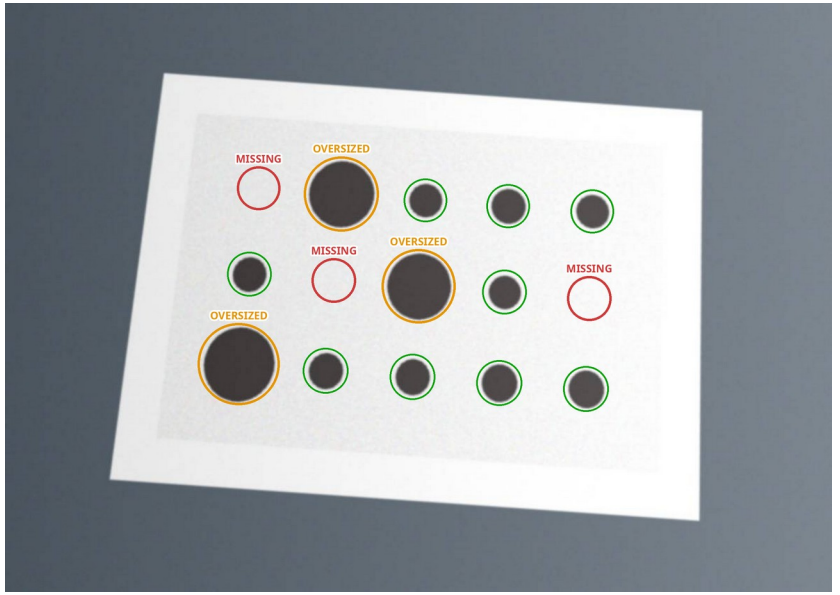
All test images are synthetic — NSW provided no production photos. Method, limits and every table: evaluation/README.md.

FEASIBILITY

Works on the photo an operator actually takes



Phone photo: card on a desk, camera tilted and off-centre



Recipe lined up (scale, rotation, perspective), then measured: 3 missing, 3 oversized, carryover p = 0.010

0% → 89%

of phone-style photos measured correctly, before → after registration — the same as perfectly framed ones

0

false alarms from zoom, framing, tilt or a desk in shot

Photo simulated from a printed sample card (the evaluation's 'on desk' condition). Validating on real NSW photos is step one of the pilot.

TRUST

Closing a case is not the same as fixing it

CONFIRMED

All five proved

- a check confirmed one cause
- exactly one change, aimed at it
- re-measured against a 'before' photo
- in spec on every check
- held 20+ shots

The only cases future rankings learn from

PLAUSIBLE

Some proved

- e.g. cause certain (tier T1) — but one deposit is still missing after the fix

Shown as context; adjusts nothing

UNVERIFIED

Nothing proved

- filed with the gaps named — e.g. '3 things changed at once: no way to say which one worked'

Adjusts nothing

Precedent can lower confidence, never raise it — so the system can't talk itself into being right.

INDUSTRY IMPACT

Ready to pilot on an NSW line

Value on day one

- A new technician's first check — ranked, with reasons
- An audit-ready report for every case, printable as PDF
- A troubleshooting history that only keeps proved fixes
- Runs offline on a line PC; a phone camera is enough

Pilot plan

1. Validate on 50–100 real NSW photos per product
2. Subtract a bare-board reference photo to remove clutter false alarms
3. Rules as data: engineers add causes (e.g. up height, NPL §3.3) without code
4. Predict: open a case automatically when a line's size spread starts drifting

Live demo next →

BACKUP · CONFIDENCE TIERS

Four confidence tiers — and agreement never raises one

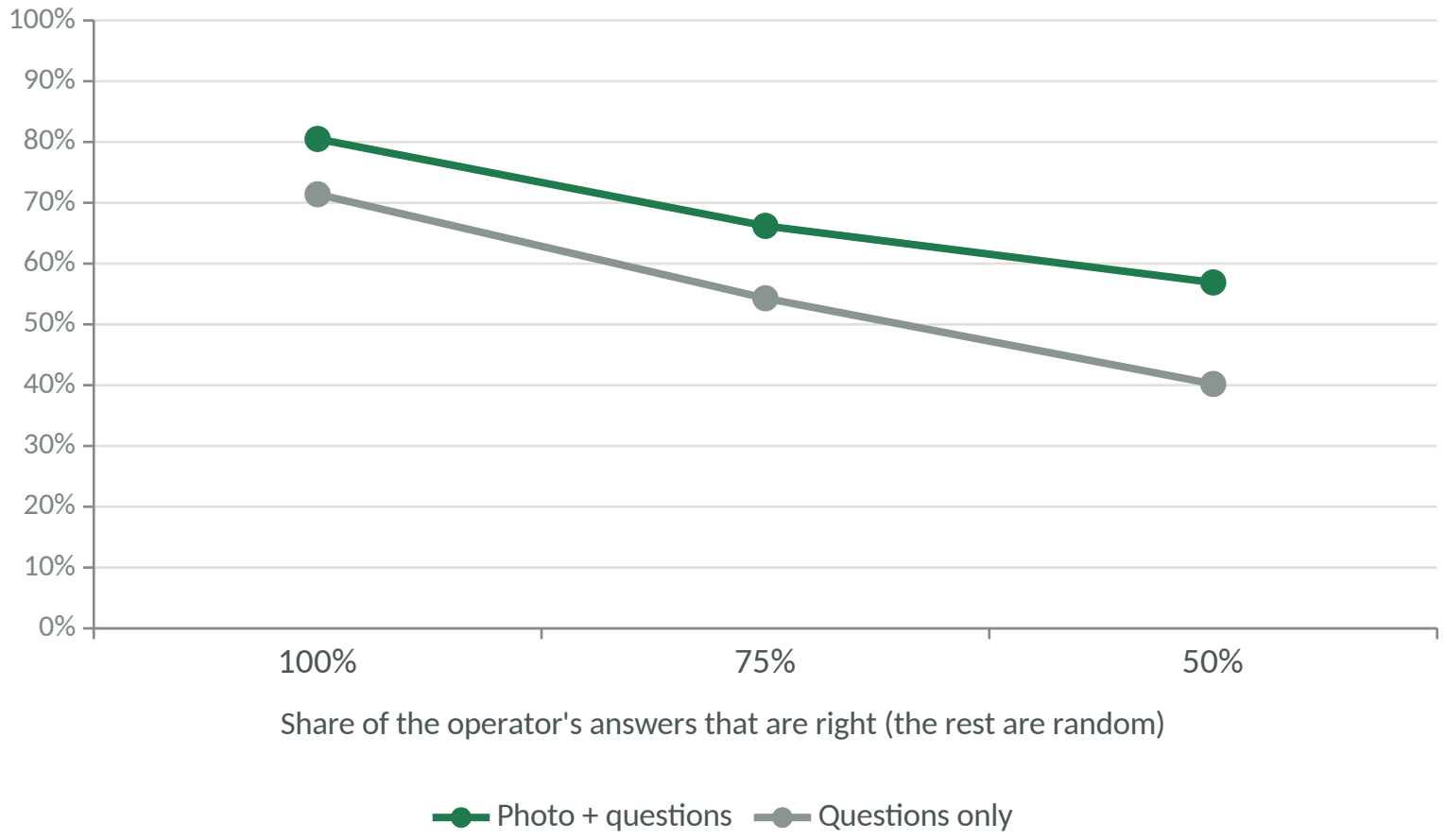
Tier	Reached when	What the operator sees
T1	Rule engine and critic agree · deciding measurement beats chance · ≥3 similar confirmed cases, none contradicting	One lead cause, with evidence and matching past cases
T2	They agree, but no precedent or the measurement isn't decisive	One lead cause, marked provisional, with what would confirm it
T3	They disagree · the measurement beats chance	Both candidates; the measurement holds the top slot
T4	They disagree · the measurement can't separate them	No lead cause — the check that would separate them

Gate A — chance. Only a signal that separates causes counts: the dispense-order pattern (binomial test, $p < 0.05$). A size spread is real but shared by several causes, so it can't decide.

Gate B — precedent. Similar confirmed cases (same material and dispenser, signals within one spread) can demote the lead by up to 0.4 log-odds — never promote it.

BACKUP · EVALUATION

The photo matters most when the answers are worst



57% vs 40%

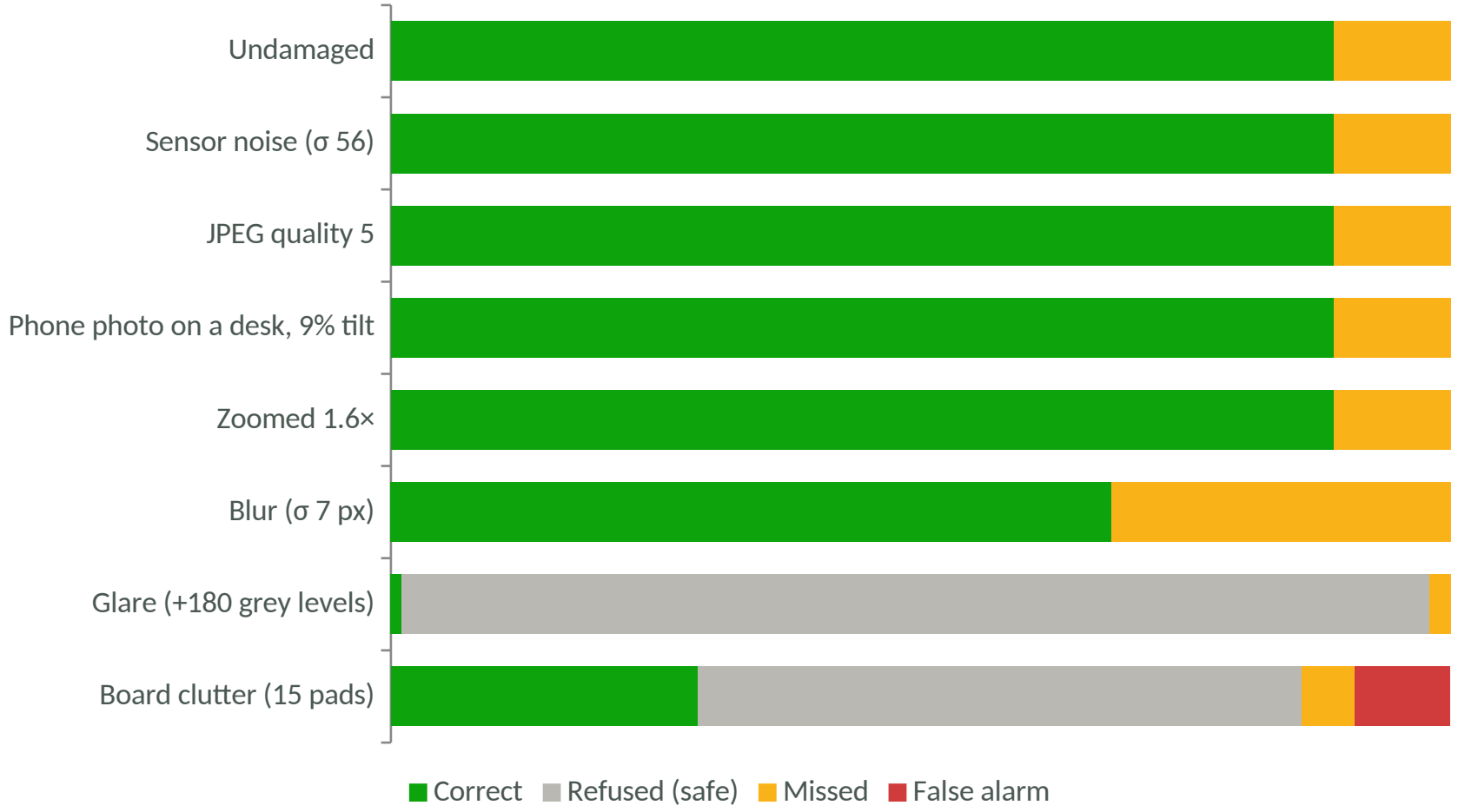
top cause right when half the answers are guesses — with the photo, and without it.

- **With the photo: 81% → 66% → 57%**
- **Questions only: 71% → 54% → 40%**

Chance is 14% (one in seven). 420 simulated cases per point; answers from scenario cards written from the NPL guide, not from the rules.

BACKUP · EVALUATION

It refuses rather than misleads



9%

false alarms under light board clutter — the only tested fault that misleads. The fix is a bare-board reference photo.

Everything else measures correctly or is refused with a reason. 'Missed' is the dispense-order pattern found in about half of carryover photos — the interview then carries it.

BACKUP · RULE c.2 DECLARATION

What we built, and what we used

Our own work

- Fault tree and its weighted, cited rules
- Two-pass ranking, chance and precedent gates, T1–T4
- The five-condition close rule
- Five-axis interview and follow-up selection
- Retrieval and precedent counting
- Photo registration, desk masking, metrology
- Synthetic defect generator and the evaluation

Third-party, declared

- NPL Report MATC(A)82 — Best Practice for Electronic Materials Dispensing (2001), © Crown copyright
- NSW Automation challenge brief
- FastAPI, Pydantic, SQLAlchemy, Alembic · OpenCV, NumPy
- Next.js, React, Tailwind, shadcn/ui, Base UI, lucide
- Groq API, openai/gpt-oss-20b — optional critic, mapping, explanation
- Dev only: pytest, Playwright, matplotlib

No model was trained or fine-tuned, and no training data was used. Every measurement is reproducible by an engineer with a microscope.

BACKUP · LIMITATIONS

Known limitations, stated up front

No real production photos

Every threshold is tuned on synthetic images; the stress test measures how far that holds, it doesn't claim real-photo accuracy.

Seven causes

Stringing, smearing and satellites (up height, NPL §3.3) have no cause yet — they're reported as 'no cause ahead' and escalated.

Busy boards

A trace touching a deposit can read as a distorted shape: 9% false alarms under light clutter.

Pixels, not microns

Absolute volume needs a scale reference; today everything is relative to the recipe.

The LLM isn't reproducible

Critic and explanation can vary between runs — which is why neither can change a likelihood.